UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Roll No. ....

Paper: Sampling Techniques (Theory) | **Course Code: STAT-305** | **Time: 3 Hrs** | **Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: **(6x5=30)**

i) Explain what do you understand by random sampling? What advantages it has over non-random sampling?

ii) Explain the concept of controlled selection.

iii) What are the points to be considered while estimating the sample size?

iv) Write a note on bias and sources of bias.

v) Intuitively Systematic Sampling is more precise than Simple Random Sampling and Stratified Sampling, discuss.

vi) Elaborate the concept of deep stratification.

Answer the following questions. **(3x10=30)**

Q.2. Enumerate all possible sample of size 2 by without replacement method from small colony of 7 households having monthly income in rupees as follows:

Household	1	2	3	4	5	6	7
Income	156	149	166	164	155	169	159

Calculate population mean $\bar{Y}$ and mean square error S^2

(a) Show that the mean gives an unbiased estimate of the population and find its sampling variance.

(b) $v(\bar{y}) = [(N-n)/2nN](y_1 - y_2)^2$, is an unbiased estimate of $V(\bar{y})$.

Q. 3 Compare the values obtained for $V(p_{st})$ under proportional allocation and optimum allocation for fixed sample size in the following two populations. Each stratum is of equal size. The f.p.c may be ignored.

Population 1		Population 2	
Stratum	P_h	**Stratum**	P_h
1	0.1	1	0.01
2	0.5	2	0.05
3	0.9	3	0.10

Q.4 Show that the variance of the mean of the systematic sample is

$$V_{(\bar{y}_{sy})} = \left(\frac{N-1}{N}\right)S^2 - \frac{k(n-1)}{N}S^2_{wsy}$$

where $S^2_{wsy} = \frac{1}{k(n-1)}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i.})^2$